

WASP-50b

R-1432

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_211020 · created 2026-08-01T07:36:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459507.863 +/- 0.061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.049
Transit depth (Rp/Rs)^2	3.05 +/- 1.72 [%]
Orbital Inclination (inc)	82.86 +/- 2.85
Airmass coefficient 1 (a1)	1.279 +/- 0.034
Airmass coefficient 2 (a2)	-0.16 +/- 0.021
Scatter in the residuals of the lightcurve fit is	2.67 %
Best Comparison Star	None
Optimal Aperture	4.08
Optimal Annulus	12.61
Transit Duration (day)	0.033 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.049 vs 0.1404 (deviation 0.71σ)
Duration fit vs published	0.033 d vs 0.0754 d (deviation 1.33σ)
Mid-time O–C	-53.7 min
Depth SNR	3.6
Residual scatter	2.67 %

Light curve

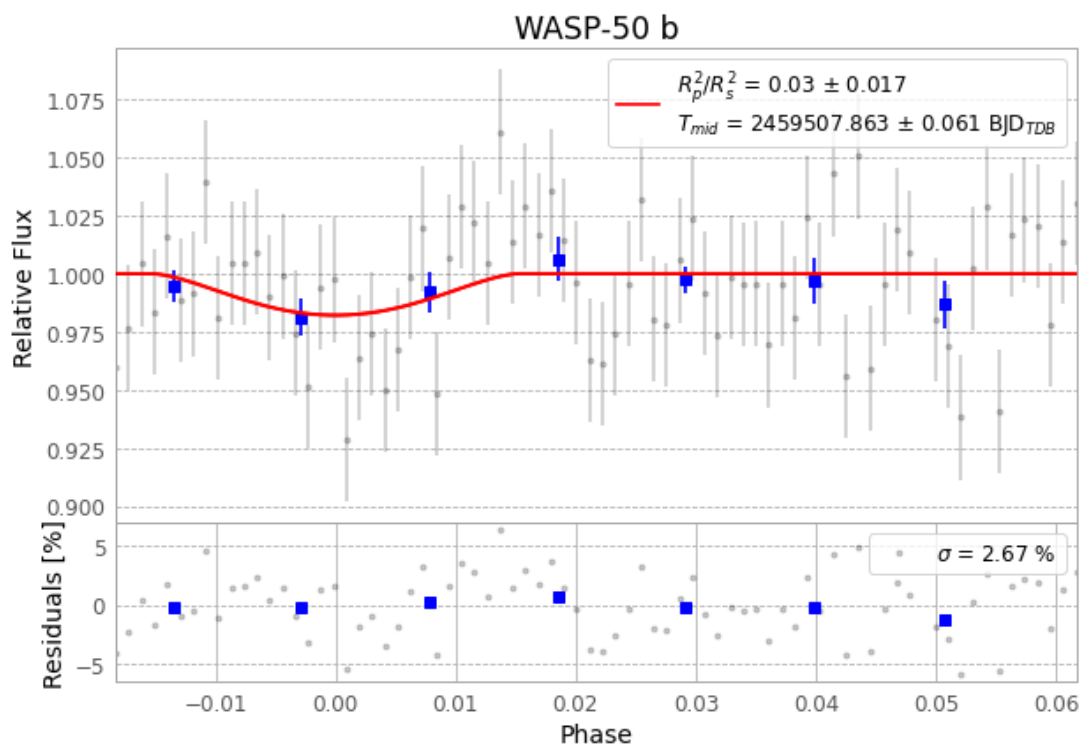


Figure 1 — FinalLightCurve_WASP-50 b_2021-10-20.png (SHA-256 c5d53525037484c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8af44feff6d59821...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-50211020074212.FITS → WASP-50211020112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-211020.log (01fb9db05ef4...), FinalLightCurve_WASP-50 b_2021-10-20.png (c5d535250374...), FinalLightCurve_WASP-50 b_2021-10-20.csv (5a3a09a7ab5d...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:36Z.